

► ENDOCRINE—ANATOMY

Pituitary gland

Anterior pituitary
(adenohypophysis)

Secretes FSH, LH, ACTH, TSH, prolactin, GH, and β -endorphin. Melanotropin (MSH) secreted from intermediate lobe of pituitary. Derived from oral ectoderm (Rathke pouch).

- α subunit—hormone subunit common to TSH, LH, FSH, and hCG.
- β subunit—determines hormone specificity.

Proopiomelanocortin (POMC) derivatives— β -endorphin, **ACTH**, and **MSH**. Go **pro** with a **BAM**!

FLAT PeG: FSH, LH, ACTH, TSH, PRL, GH.

B-FLAT: Basophils—FSH, LH, ACTH, TSH.

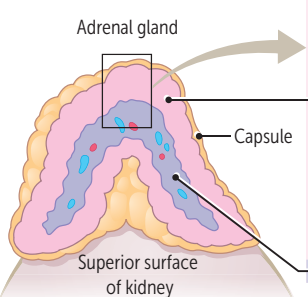
Acid PiG: Acidophils — PRL, GH.

Posterior pituitary
(neurohypophysis)

Stores and releases vasopressin (antidiuretic hormone, or ADH) and oxytocin, both made in the hypothalamus (supraoptic and paraventricular nuclei) and transported to posterior pituitary via neurophysins (carrier proteins). Derived from **neuro**ectoderm.

Adrenal cortex and medulla

Adrenal cortex (derived from mesoderm) and medulla (derived from neural crest).

ANATOMY	HISTOLOGY	1° REGULATION BY	HORMONE CLASS	1° HORMONE PRODUCED
	Zona G lomerulosa	Angiotensin II	Mineralocorticoids	Aldosterone
	Zona F asciculata	ACTH, CRH	Glucocorticoids	Cortisol
	Zona R eticularis	ACTH, CRH	Androgens	DHEA
	MEDULLA Chromaffin cells	Preganglionic sympathetic fibers	Catecholamines	Epi, NE

GFR corresponds with **s**alt (mineralocorticoids), **s**ugar (glucocorticoids), and **s**ex (androgens).

Endocrine pancreas cell types

Islets of Langerhans are collections of α , β , and δ endocrine cells. Islets arise from pancreatic buds.

α = gluc **α** gon (peripheral)

β = insulin (central)

δ = somatostatin (interspersed)

